

Latch remembers what must happen.

MementoVM turns future commitments into typed, versioned programs that act once - and forget on purpose.



History is not a plan for the future

Vector memory

Retrieves what happened. Similarity can surface stale plans, but it does not encode inhibitors, deadlines, approval policy, or an exactly-once lifecycle.

Prospective memory

Compiles what must happen. Typed cues can be revised, inhibited, deferred, approved, executed once, and removed from hot memory after their purpose is complete.

The gap is a control problem - not a retrieval problem.

Intent becomes a durable program

Type intent. Inspect every decision.

Qwen compiles natural language into an Intention Program. The Memory Debugger exposes versions, predicates, rejected distractors, approval state, and the exact evidence behind action or non-action.

The screenshot displays the Latch interface, which is used for managing and monitoring intention programs. The interface is divided into several sections:

- Header:** Includes the Latch logo, a navigation bar with "Latch" and "Overview", and a "DETERMINISTIC DEMO" button.
- Left Sidebar:** Contains navigation links for "Avery's workspace", "Overview", "Chat", "Active memory", "Event timeline", "Simulator", "Evaluation", and "System".
- Prospective Memory Runtime:** A section titled "Remember what must happen next." with a description: "Latch turns future commitments into typed, versioned programs—then notices the right cue, stays quiet on distractors, and acts once under your control." It includes a "Run the proof" button.
- Active Programs:** A section showing "1" active program, "Bounded hot-memory set".
- Candidates Retrieved:** A section showing "0" candidates, "From the latest event".
- PM F1:** A section showing "100%" performance, "Measured on PM-Mini v1".
- Duplicate Actions:** A section showing "0" duplicate actions, "Exactly-once benchmark".
- Active Intention Program - V2:** A section titled "Prepare Dana's DPA redline" with a "DORMANT" status. It includes a description: "Also wait for finance approval before preparing it." and a flowchart showing the sequence of actions: "Legal approval" (contract 043 - v7) → "Finance approval" (deal 043) → "Prepare draft" (Dana - approval required). Below the flowchart are three status indicators: "Inhibitor: contract stays open", "Focus block: 09:00-11:00", and "Idempotency: once".
- LIVE PROOF:** A section titled "Next simulator step" with a "10 steps" indicator. It includes a "Marketing approval lure" with a description: "Related wording, wrong channel and entity."

A deterministic runtime surrounds Qwen

01 Chat + events

One normalized contract for users, connectors, and simulator.

02 Qwen compiler

Structured JSON only; thinking disabled; one repair pass.

03 Schema + policy

Types, tools, risk, and approval rules validated locally.

04 PostgreSQL

Versioned programs, cue indexes, jobs, approvals, and audit.

05 Exact cue router

Channel, event type, entity, validity, and inhibitor first.

06 Worker + clock

Time and absence monitoring with deterministic demo time.

07 Approval + executor

Action hash, expiry, atomic claim, idempotency key.

08 Alibaba OSS

Private replay bundles and deployment proof artifacts.

Safety is a database invariant, not a prompt

Typed, immutable programs

Every material revision creates a complete version; the prior version is superseded and removed from hot retrieval.

Exact before semantic

Hard IDs, validity, and inhibitors resolve locally. Only bounded ambiguity is eligible for Qwen adjudication.

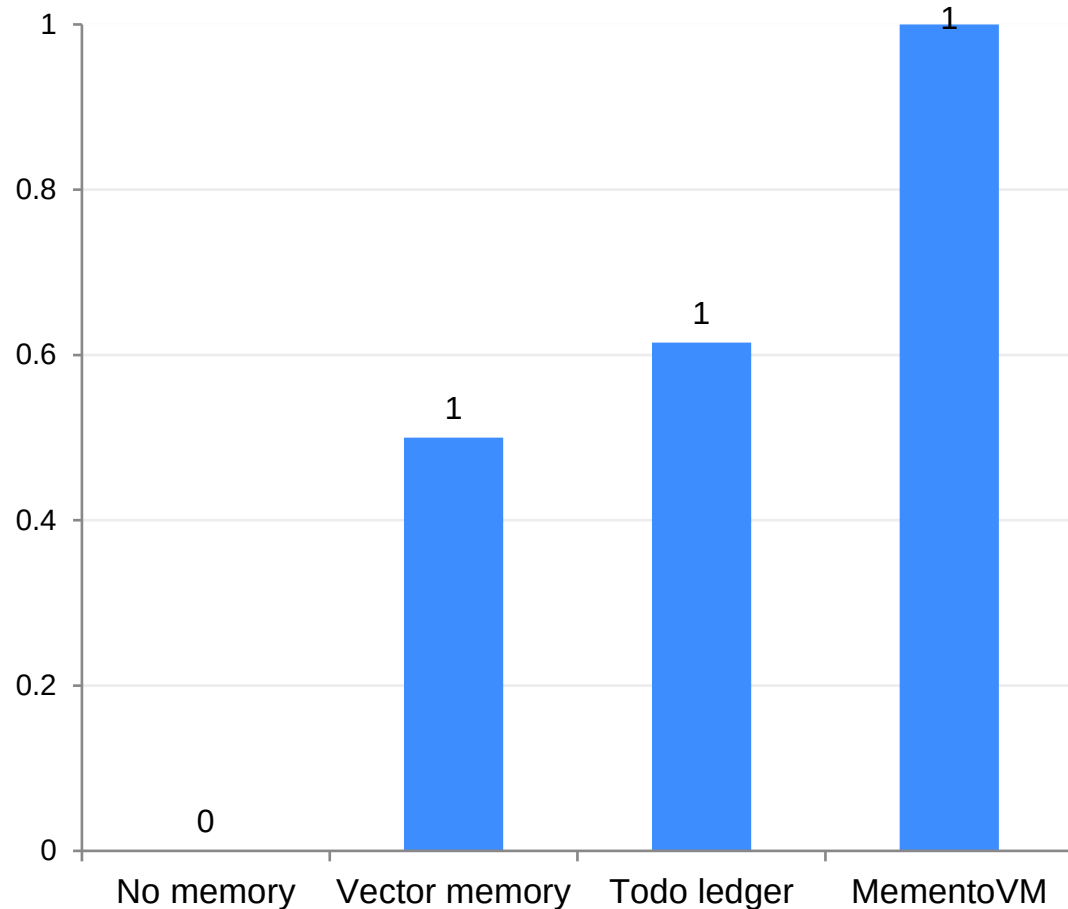
Approval bound to one action

External writes require a human decision tied to the version, action hash, and a 30-minute expiry.

Exactly once under retries

A database claim and unique idempotency key suppress duplicate webhooks, clicks, workers, and tool retries.

MementoVM reaches 1.00 PM F1 on PM-Mini



60

original deterministic scenarios across exact cues, lures, stale state, inhibitors, absence, and cancellation.

0

false alarms, misses, and duplicate actions for MementoVM on PM-Mini v1.

10/10

complete judge-scenario runs passed against the live local container stack.

Synthetic release dataset; results characterize PM-Mini v1, not general production performance.

Prospective memory is the missing control plane.

github.com/1aifanatic/mementovm

60-case benchmark • deterministic simulator

Qwen Cloud • Alibaba Cloud deployment profile

